DataManager

System Description

Abstract

This document describes the DataManager support system of the Eclipse Arrowhead 5.2.2 framework, as implemented in `ah5-DataManager-java-spring` of the primary Java source tree (`Eclipse-Arrowhead-java`) and deployed with the standalone Compose definition of that folder.

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1 Overview

This document describes the DataManager system. DataManager caches and persists SenML (RFC 8428) sensor and actuator data: `dataProxy` keeps the latest message per system and service, `dataHistorian` the time series.



Figure 1: SysML-style block view of DataManager with its produced and consumed services.

The rest of this document is organized as follows. In Section 1.1, we reference major prior art capabilities of the system. In Section 1.2, we describe the intended usage of the system. In Section 1.3, we describe fundamental properties provided by the system. In Section 1.4, we describe delimitations of capabilities of the system. In Section 2, we describe the abstract service functions consumed or produced by the system. In Section 3, we describe the security capabilities of the system.

1.1 Significant Prior Art

Port of the Arrowhead 4.6 DataManager to the 5.2.2 baseline.

1.2 How This System Is Meant to Be Used

Producers `put-data`; consumers `get-data` and browse with `list-systems` and `list-services`. In the earlier combined test-cloud deployments the producers also forwarded data across clouds through the Gatekeeper.

1.3 System functionalities and properties

1.3.1 Functional properties of the system

- SenML validation and completion.
- Proxy cache of the last message.
- Historian storage and query by time, count and signals.

1.3.2 Configuration of system properties

The system is configured with Spring properties (`application.properties`, overridden by environment variables in the container deployment). The properties that shape its behaviour are:

Property	Default	Meaning
`server.port`	8461	HTTPS port.
`authentication.policy`	declared	Standalone deployment: `outsourced` with Authentication identity tokens and mutual TLS.
`management.policy` / `management.whitelist`	authorization	Access to the management services.
`enable.blacklist.filter` / `force.blacklist.filter`	false / true	Blacklist check of requesters.
`service.authorization.enabled`	false	Per-call ConsumerAuthorization check (`authorization.verify`, cached 30 s, fail closed); standalone deployment: true.
`normalization.mode`	extended	Name normalization.
`sysmonitor.enabled`, `sysmonitor.broker.*`, `sysmonitor.ssl.*`	true, amqp-broker:5671, server.ssl stores	Native `sysMonitor` publisher of the common library.
`arrowhead.ssl.crl-file`	(unset)	CRL checked for every client certificate (standalone deployment: `/crl/ca.crl`).

Secrets (database, keystore and identity passwords, secret keys) have no usable defaults; the standalone deployment reads them from per-deployment Compose file secrets under `/run/secrets`.

The standalone deployment of this folder uses `config/application.properties` (copied into the image) with `server.port`=8461, `authentication.policy=outsourced`, `service.interface.policy=CERT_AUTH`, `management.policy=sysop-only`, `service.authorization.enabled=true`, `mqtt.api.enabled=false`, `arrowhead.ssl.crl-file=/crl/ca.crl`; secrets are read from Compose file secrets under `/run/secrets` (`spring.config.import=configtree:/run/secrets/`). `enable.blacklist.filter` is not enabled there, so incoming requests are not checked against the Blacklist; blacklisted providers are still removed by DynamicServiceOrchestration.

1.3.3 Data stored by the system

Historian records and log entries (database `ah_datamanager`); the proxy cache is in memory.

1.3.4 Non functional properties

`echo` is `degraded` while the historian database cannot be queried.

1.3.5 Stateful or stateless

Stateful: historian records are persistent; the proxy cache is lost at restart.

1.4 Important Delimitations

No retention policy for the historian.

2 Services

This section describes the produced and consumed services. Each produced service is specified by an SD document and one IDD document per interface.

2.1 Produced services

Service	Version	Interfaces	SD	IDD
`dataProxy`	1.0.0	HTTP(S)	[1]	[2]
`dataHistorian`	1.0.0	HTTP(S)	[3]	[4]
`sysMonitor`	1.0.0	AMQP(S)	[5]	[6]

The HTTP interfaces are always offered and the AMQP `sysMonitor` interface with `sysmonitor.enabled=true`.

2.2 Consumed services

Provider	Service	Operations	Use	Condition
ServiceRegistry	systemDiscovery, serviceDiscovery	register, revoke, lookup	Registration of the system and its services; lookup of the bootstrap services.	always
Authentication	identity	identity-login, identity-verify	Own identity token; verification of requester tokens.	`authentication.policy=outsourced`
Consumer Authorization	authorization	verify	Per-call authorization of its services.	`service.authorization.enabled=true`
Consumer Authorization	authorization	verify	Management filter decisions.	`management.policy=authorization`
AMQP broker (SysMonitor)	sysMonitor (publisher)	echo, power, status	Own observations.	`sysmonitor.enabled=true`

Except for the bootstrap services, which are looked up in the ServiceRegistry, the system reaches other systems' services through DynamicServiceOrchestration (`http.collector.mode=ORCHESTRATION`, results cached for 300 s); every such call needs a ConsumerAuthorization grant.

3 Security

The system can run in an insecure mode (`server.ssl.enabled=false`, plain HTTP, identity declared by the requester) for development, and in the Arrowhead secure mode with HTTPS and X.509 certificates following the Arrowhead certificate profile (`<System>.<Cloud>.<Organization>.arrowhead.eu`). Certificates not issued under the cloud CA are refused by the TLS handshake. The standalone deployment of this folder (`config/application.properties` and `compose.yaml`) runs the secure mode.

3.1 Security model

- **Protocols:** HTTPS (TLS 1.3 and 1.2) with mandatory client certificates; MQTT over TLS; AMQP over TLS for `sysMonitor`.
- **Data protection:** transport encryption by TLS; the database connection uses TLS with server verification in the standalone deployment. Secrets are kept out of the container environment (Compose file secrets).
- **System authentication:** by the authentication policy: `certificate`, `outsourced` (identity token verified with Authentication) or `declared` (insecure); the standalone deployment uses `outsourced`. An identity token is accepted only together with the client certificate of the system it was issued to (token-to-certificate binding), and every client certificate is checked against the cloud CA's CRL (fail closed).
- **Authorization of produced services:** management services are protected by the management filter (`management.policy`: `SYSOP_ONLY`, `WHITELIST` or `AUTHORIZATION`); with `service.authorization.enabled=true` ConsumerAuthorization decides each call; an optional blacklist filter refuses blacklisted requesters. Sysop identities bypass the management filter.
- With service authorization every call must come from a client certificate matching the authenticated system, and ConsumerAuthorization must permit the operation.

For the Arrowhead certificate profile see github.com/eclipse-arrowhead/documentation.

4 References

[1] J. Delsing, "dataProxy – Service Description (DataManager)," Eclipse Arrowhead, SD 5.2.2, 2026, dataManager-v5.2.2-dataProxy-SD.pdf.

[2] ——, "dataProxy HTTP/TLS/JSON – Interface Design Description (DataManager)," Eclipse Arrowhead, IDD 5.2.2, 2026, dataManager-v5.2.2-dataProxy-HTTP-IDD.pdf.

[3] ——, "dataHistorian – Service Description (DataManager)," Eclipse Arrowhead, SD 5.2.2, 2026, dataManager-v5.2.2-dataHistorian-SD.pdf.

[4] ——, "dataHistorian HTTP/TLS/JSON – Interface Design Description (DataManager)," Eclipse Arrowhead, IDD 5.2.2, 2026, dataManager-v5.2.2-dataHistorian-HTTP-IDD.pdf.

[5] ——, "sysMonitor – Service Description (DataManager)," Eclipse Arrowhead, SD 5.2.2, 2026, dataManager-v5.2.2-sysMonitor-SD.pdf.

[6] ——, "sysMonitor AMQP/TLS/JSON – Interface Design Description (DataManager)," Eclipse Arrowhead, IDD 5.2.2, 2026, dataManager-v5.2.2-sysMonitor-AMQP-IDD.pdf.

5 Revision History

5.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-22	5.2.2	Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace.	Jerker Delsing
2	2026-10-30	5.2.2	Checked against the primary Java source tree (Eclipse-Arrowhead-java): standalone deployment, configuration and security statements updated.	Claude Opus 5.5 (draft)

5.2 Quality Assurance

No.	Date	Version	Approved by
1		5.2.2	